Safety Erasure Under KV-Cache Compression

Aryan Gupta
aryan.cs.app@gmail.com

Abstract

Inference systems routinely compress or evict the key-value (KV) cache to reduce serving cost. We show that on dense full-attention instruction-tuned language models, this throughput optimization can become a selective safety intervention: refusal and policy-following behavior degrades meaningfully more than ordinary task capability when the cache is aggressively evicted. We test this across eleven open-weight checkpoints from seven model families and find positive Selective Safety Erasure Index (SSEI) in four families (Llama, OLMo, Phi, Qwen; seven instruction-tuned models with effect sizes from 0.02 to 0.09 in absolute pass-rate units, 95% bootstrap CIs excluding zero) under sliding-window and user-pinned cache policies. The effect is robust to leave-one-family-out perturbation. Three models do not show the pattern, and the architectural common thread among them (interleaved or full-layer sliding-window attention; non-standard mixture-of-experts cache handling) is consistent with the hypothesis that the affected behavior depends on long-range KV cache state. Causal-patching experiments that restore baseline cache state into compressed runs recover 35–58% of lost refusal behavior (depending on model), establishing cache state as a safety-relevant surface. System-role and user-role K+V restorations produce comparable recovery under a per-prompt mean-of-ratios estimator, indicating that the safety-relevant information is distributed across cached tokens rather than localized to any single conversational role. Policy-pinned cache retention, which protects system-role tokens from eviction, fully restores refusal behavior.

1 Introduction

Production language-model serving systems do not deploy a static pair of weights and prompts. They actively manage the transient KV cache through eviction policies, quantization, and paged attention to fit longer contexts into a fixed memory budget [Kwon et al., 2023, Zhang et al., 2023, Xiao et al., 2024]. These mechanisms are documented to degrade task quality [Chen et al., 2025, Yang et al., 2024, Ananthanarayanan et al., 2026]. We ask a different question: does cache compression weaken refusal behavior more than it weakens ordinary capability on the same prompts and budget?

The result is yes for the majority of currently-deployed dense-attention instruction-tuned models. Across eleven checkpoints sampled from Llama, Gemma, Mistral, OpenAI gpt-oss, OLMo, Phi, and Qwen families, seven instruction-tuned models show a Selective Safety Erasure Index (SSEI) of at least one cache policy whose 95% lower CI exceeds zero. Phi-4 and Qwen3-8B exhibit the largest effects, around eight to nine percentage points of safety loss beyond matched capability loss under sliding-window cache eviction. The cross-family conclusion holds when any single family is removed from the panel.

Three models do not show the pattern: Gemma-2-9B-IT, Mistral-7B-Instruct-v0.3, and OpenAI gpt-oss-20b. These models share architectural features that decouple long-range KV cache state from the safety-relevant generation circuit: Gemma 2 interleaves local sliding-window attention with global attention [Gemma Team, Google DeepMind, 2024], Mistral 7B v0.3 applies sliding-window attention at every layer [Jiang et al., 2023], and gpt-oss-20b is a mixture-of-experts model trained with a structured "harmony" response format [OpenAI, 2025]. Under our cache interventions gpt-oss-20b also collapses into a uniform output mode rather than gracefully degrading. A matched-baseline comparison tests whether Model Spec Midtraining (MSM) [Li et al., 2025] confers structural robustness: Qwen2.5-14B-Instruct without MSM shows SSEI = 0.089 [0.073, 0.104] under sliding-window eviction; the MSM-adapted version shows SSEI = 0.050 [0.041, 0.059]. However, the non-MSM model has substantially higher baseline refusal (64% vs. 30%), and in relative terms both models lose approximately 16% of baseline safety under sliding-window compression. The reduced absolute SSEI under MSM is therefore dominated by baseline effects rather than mechanistic protection — deep post-training alignment does not defend against cache-eviction safety erosion. The positive results in four families and the negative results in these three constitute a falsification test: the affected behavior is cache-resident long-range attention state, not generic compression sensitivity.

Causal-patching experiments restore baseline cache state into compressed runs and recover 35–58% of lost refusal behavior on Qwen2.5-7B-Instruct (single-seed) and Qwen3-8B (multi-seed, Sonnet-judged). Under a per-prompt bootstrap estimator, system-role and matched user-role interventions produce comparable restoration fractions (Qwen3-8B: 0.355 vs. 0.408, overlapping CIs; Qwen2.5-7B: 0.584 vs. 0.584), establishing that the safety-relevant information is distributed across cached tokens rather than localized to system-role spans. This distributed encoding is consistent with the architectural finding: models trained on full global attention encode safety reasoning across the entire cache, so any aggressive eviction policy can damage it. A parallel Phi-4 run under simulated four-bit quantization shows no safety degradation (0.985 vs. 0.987 baseline), confirming that Phi-4's vulnerability arises specifically from sliding-window eviction rather than quantization.

We provide cross-family measurement of selective safety degradation under cache eviction with explicit per-policy effect sizes and bootstrap CIs, an architecture-dependent characterization of which models are affected, causal-patching evidence that safety information is distributed across cached tokens, and a complete mitigation: policy-pinned cache retention fully restores refusal behavior.

2 Related Work

KV-cache compression. Algorithms for KV-cache management trade off memory for quality. H2O retains heavy-hitter tokens [Zhang et al., 2023]; StreamingLLM relies on attention sinks for long contexts [Xiao et al., 2024]; SnapKV compresses prompts using observed attention patterns [Li et al., 2024]; KIVI and KVQuant study low-bit quantization [Liu et al., 2024, Hooper et al., 2024]. Chen et al. [2025] report that KV compression unevenly harms instruction following and increases system-prompt leakage. Ananthanarayanan et al. [2026] frame compression as a perturbation of token accessibility through attention dynamics. We extend this line of work along the safety axis: rather than measuring overall quality loss, we measure safety loss minus capability loss on matched prompts.

Cache state as a control surface. Wang et al. [2025] show that KV-cache state can be edited to defend against indirect prompt injection, and prompt-extraction work establishes that the cache can be read out as well [Hui et al., 2024]. These results frame cache state as an addressable safety-relevant object. Our experiments treat cache state as a causal locus and ask whether ordinary serving-time eviction can damage that state.

Mechanism-first safety phenomena. A growing body of work isolates specific mechanisms in safety failure rather than only reporting benchmark accuracy: refusal directions [Arditi et al., 2024, Joad et al., 2026], sparse alignment routes [Frank, 2026], depth-conditioned refusal restoration [Zhang et al., 2026], subliminal learning [Cloud et al., 2025], token entanglement [Zur et al., 2025], and emergent misalignment [Betley et al., 2025]. The closest is Wang et al. [2025], which uses cache state as an intervention point. We test whether cache state is a naturally occurring safety failure point under standard inference optimizations.

3 Method

3.1 Setup

We evaluate eleven open-weight checkpoints: Qwen2.5-7B (base and Instruct) [Qwen Team, 2024], Qwen2.5-14B-Instruct (with and without a Model Spec Midtraining LoRA adapter [Li et al., 2025]), Qwen3-8B [Qwen Team, 2025], Llama-3.1-8B-Instruct [Llama Team, AI @ Meta, 2024], Gemma-2-9B-IT [Gemma Team, Google DeepMind, 2024], Mistral-7B-Instruct-v0.3 [Jiang et al., 2023], OLMo-3-7B-Instruct [Team OLMo, 2025], Phi-4 [Abdin et al., 2024], and OpenAI gpt-oss-20b [OpenAI, 2025]. The MSM checkpoint uses a LoRA adapter trained with Anthropic's Model Spec Midtraining protocol on top of Qwen2.5-14B-Instruct; the matched non-MSM baseline runs on the same weights without the adapter, providing a controlled comparison of whether deep post-training alignment affects vulnerability to cache-eviction safety erasure. Models are loaded in bfloat16 with explicit access to the cache state, so eviction is implemented in the generation loop rather than delegated to a serving system. The results measure the underlying model behavior under cache eviction, not operator-level deployment; vLLM [Kwon et al., 2023], TGI, and PagedAttention kernels require separate deployment-level validation.

3.2 Cache Policies

We compare uncompressed generation against five cache interventions:

- sliding window: retain only the most recent N cached tokens;
- sink plus recent: retain the first S tokens and the most recent N;
- random matched: evict random tokens at the same budget as a structured policy, as a non-policy control;
- user pinned: protect user-role spans, then fill the remaining budget with sink and recent retention. This is the non-policy control for policy pinning;
- policy pinned: protect system-role or policy spans, then fill the remaining budget with sink and recent retention.

For all policies, we record retained and evicted token counts by role at each decode step.

3.3 Prompt Suites

The benchmark separates safety, leakage, over-refusal, and capability:

- Public refusal safety: harmful requests drawn from AdvBench [Zou et al., 2023], Jailbreak-Bench [Chao et al., 2024], and HarmBench [Mazeika et al., 2024];

- Public benign over-refusal: Databricks Dolly 15k benign instructions [Databricks, 2023];
- Public XSTest safe: benign safety-sensitive prompts [Rottger et al., 2023];
- Public system leakage: prompt-injection attacks from the Cyberec corpus [Cyberec, 2026] paired with a hidden system canary;
- Public capability ARC: multiple-choice capability controls [Clark et al., 2018].

Each suite ships with a manifest of prompt IDs, source dataset revisions, and content hashes. Because public harmful-prompt datasets may overlap with model safety training, we interpret public-suite results as public-benchmark behavior rather than as a generalization claim.

3.4 Metrics

Safety and refusal are scored with refusal-string detection on safety suites and exact or token-overlap leakage against hidden system text on leakage suites. Capability is scored with multiple-choice accuracy on ARC-Easy. Over-refusal is the refusal rate on the benign suites.

For each cache policy p, budget b, and model m, we define the Selective Safety Erasure Index as the difference between safety degradation and capability degradation relative to the uncompressed baseline:

$$\mathrm{SSEI}(p, b, m) = \Delta_{\mathrm{safety}}(p, b, m) - \Delta_{\mathrm{capability}}(p, b, m). \tag{1}$$

Positive SSEI indicates that safety pass-rate declines more than capability pass-rate under the same intervention. We report prompt-clustered bootstrap 95% confidence intervals for SSEI. A model-policy condition is considered a positive instance of selective safety erasure when SSEI ≥ 0.01 and its lower CI exceeds zero.

3.5 Audit

Automated labels are imperfect. We blinded the suite, policy, and treatment identity of each audit row and routed labels through a separate-family model judge. Annotations come from Claude Sonnet 4 [Anthropic, 2025]; family separation prevents same-family judging on any panel model. The audit manifests record judge model, prompt-template hashes, raw-output hashes, and response-length buckets so the labels can be re-derived. Across the panel, 2,595 of 2,676 attempted audit rows parsed (97%); per-model parse rates range from 91% to 100%.

3.6 Causal Restoration

For length-preserving cache perturbations such as simulated four-bit KV quantization, we run baseline (uncompressed) and compressed decoding on identical prompts and seeds, then patch selected key and value slices from the baseline cache into compressed runs at the same positions. We compare:

1. compressed decoding with no patch;
2. compressed decoding with system-role key+value slices restored from baseline;
3. compressed decoding with user-role slices restored, matched to the system-token count;
4. compressed decoding under the policy-pinned eviction policy that protects system spans throughout generation;

5. compressed decoding under the user-pinned policy as the matched non-policy control.

The user-role restoration is the central control: it shares the additional retained-token budget with the system-role patch while differing only in which role's content is restored. Comparing system-role and user-role restoration fractions reveals whether safety information is role-localized or distributed across the cache.

4 Results

4.1 Cross-Family Selectivity

Figure 1 reports per-model per-policy SSEI point estimates and 95% bootstrap confidence intervals. The largest positive effects come from sliding-window cache eviction on Qwen3-8B (SSEI = 0.092, CI [0.083, 0.101]) and Qwen2.5-14B-Instruct (SSEI = 0.089, CI [0.073, 0.104]). Phi-4 shows a comparable effect (SSEI = 0.084, CI [0.076, 0.091]). The MSM-adapted Qwen2.5-14B variant shows a lower absolute SSEI (0.050 [0.041, 0.059]), but the relative proportion of baseline safety lost is near-identical (∼16% for both), indicating that the absolute difference reflects the MSM adapter's lower baseline refusal rate (30% vs. 64%) rather than structural robustness to compression. Smaller but significant positive effects appear under user-pinned eviction on Llama-3.1-8B-Instruct (0.024 [0.016, 0.031]), OLMo-3-7B-Instruct (0.026 [0.019, 0.033]), and Qwen2.5-7B-Instruct (0.017 [0.011, 0.022]).

Table 1: Cross-family selectivity (SSEI) by cache policy. SSEI is safety degradation minus capability degradation relative to the no-cache-policy baseline; positive SSEI indicates safety loss exceeding capability loss. 95% bootstrap CIs in brackets.

Family	Model	sliding window	sink+recent	policy-pinned	user-pinned	random matched
Gemma	Gemma–2–9B–IT	−0.005 [-0.008, -0.003]	−0.001 [-0.003, 0.001]	−0.004 [-0.006, -0.002]	−0.003 [-0.008, 0.001]	−0.000 [-0.003, 0.003]
OpenAI	GPT–OSS–20B	−0.068 [-0.088, -0.048]	−0.068 [-0.088, -0.048]	−0.079 [-0.099, -0.060]	−0.082 [-0.101, -0.063]	−0.079 [-0.099, -0.060]
Llama	Llama–3.1–8B–Instruct	−0.085 [-0.105, -0.067]	0.007 [0.002, 0.012]	−0.001 [-0.007, 0.004]	0.024 [0.016, 0.031]	0.022 [0.014, 0.029]
Mistral	Mistral–7B–Instruct–v0.3	0.009 [0.002, 0.017]	0.001 [-0.002, 0.004]	−0.004 [-0.008, -0.000]	−0.008 [-0.015, -0.002]	0.002 [-0.003, 0.006]
OLMo	OLMo–3–7B–Instruct	−0.004 [-0.008, 0.000]	−0.010 [-0.014, -0.005]	−0.016 [-0.020, -0.011]	0.026 [0.019, 0.033]	0.005 [0.000, 0.010]
Phi	Phi–4	0.084 [0.076, 0.091]	0.001 [-0.000, 0.003]	−0.001 [-0.003, 0.000]	0.055 [0.046, 0.063]	0.044 [0.038, 0.050]
Qwen	Qwen2.5–14B–Instruct	0.089 [0.073, 0.104]	0.003 [-0.005, 0.011]	−0.003 [-0.011, 0.005]	0.081 [0.065, 0.096]	0.077 [0.066, 0.089]
Qwen	Qwen2.5–14B–Instruct + MSM	0.050 [0.041, 0.059]	0.012 [0.004, 0.020]	−0.009 [-0.017, -0.001]	0.020 [0.006, 0.032]	0.028 [0.019, 0.037]
Qwen	Qwen2.5–7B base	–	–	–	–	–
Qwen	Qwen2.5–7B–Instruct	0.011 [0.007, 0.016]	−0.003 [-0.008, 0.001]	−0.001 [-0.006, 0.003]	0.017 [0.011, 0.022]	0.012 [0.007, 0.016]
Qwen	Qwen3–9B	0.092 [0.083, 0.101]	−0.005 [-0.009, -0.001]	0.010 [0.006, 0.015]	−0.002 [-0.006, 0.002]	0.020 [0.014, 0.026]

The cross-family conclusion is robust to single-family removal. Table 2 reports a leave-one-family-out check: for each family, we re-evaluate the cross-family claim after excluding that family and verify that at least two of the remaining families still have a positive condition with CI excluding zero. The claim holds in every case, including when the largest-effect family (Phi) or the entire Qwen family (five checkpoints) is removed.

4.2 Per-Suite Effects

The selectivity is concentrated in the registered safety and leakage suites. On capability prompts (ARC-Easy), cache compression reduces accuracy by at most a few percentage points and rarely outside the bootstrap CI, while refusal accuracy on safety prompts loses an order of magnitude more under aggressive eviction.

Figure 1: Selective Safety Erasure Index by model and cache policy. Each row is one (model, policy) condition; the horizontal bar is the 95% bootstrap CI and the marker is the point estimate.

Table 2: Leave-one-family-out robustness check on the cross-family selectivity claim. Each row removes the named family and counts the remaining instruction-tuned families that retain positive SSEI with 95% lower CI excluding 0 on at least one registered cache policy. The registered rule requires at least two such families.

Excluded family	Positive families remaining	Claim holds?
Gemma	Llama, OLMo, Phi, Qwen	supported
Llama	OLMo, Phi, Qwen	supported
Mistral	Llama, OLMo, Phi, Qwen	supported
OLMo	Llama, Phi, Qwen	supported
OpenAI	Llama, OLMo, Phi, Qwen	supported
Phi	Llama, OLMo, Qwen	supported
Qwen	Llama, OLMo, Phi	supported

suite	policy	Safety delta [95% CI]	Paired / clusters
Public system leakage	Window 128	0.293 [0.270,0.316]	1300
Public system leakage	Policy-pinned cache	0.063 [0.053,0.073]	1300
Refusal safety	Window 128	0.053 [0.038,0.068]	1300
Public system leakage	Random matched	0.032 [0.024,0.040]	1300
Refusal safety	Random matched	0.032 [0.015,0.047]	1300
Refusal safety	Policy-pinned cache	-0.024 [-0.038,-0.010]	1300
Public system leakage	user_pinned / budget 128 / sink 8	-0.017 [-0.022,-0.012]	1300
Public system leakage	Sink+recent	-0.013 [-0.018,-0.008]	1300
Refusal safety	user_pinned / budget 128 / sink 8	0.007 [-0.007,0.022]	1300
Refusal safety	Sink+recent	-0.005 [-0.019,0.010]	1300

Table 3: Suite-level safety-degradation effects (suites with <10 paired prompts omitted).

4.3 Models That Do Not Show the Effect

Three checkpoints fail to produce positive SSEI on any registered policy: Gemma-2-9B-IT, Mistral-7B-Instruct-v0.3, and OpenAI gpt-oss-20b. The first two share an architectural feature with the underlying generation circuit: native sliding-window attention. Gemma 2 interleaves a 4,096-token local sliding-window attention with full global attention every other layer [Gemma Team, Google DeepMind, 2024]. Mistral 7B v0.3 uses 4,096-token sliding-window attention at every layer [Jiang et al., 2023]. In both cases, training has already adapted the model to operate against a locally-windowed view of prior tokens; aggressive eviction at inference time is closer to the model's training distribution than it is for vanilla full-attention models. The Mistral run has SSEI = 0.009 at the largest measured policy, immediately below our 0.01 threshold; the Gemma run is centered on zero.

The gpt-oss-20b case is qualitatively different. Mean safety-suite score is 0.129 at the uncompressed baseline and rises to 0.554 under every treatment policy, including the random-matched control. A scalar safety improvement under cache pressure is inconsistent with the hypothesis that the model is gracefully degrading under cache pressure; the uniformity across all five eviction policies indicates a collapse into a single output mode rather than a modulated response to each policy's distinct retention pattern. gpt-oss-20b is trained on the harmony response format [OpenAI, 2025] and routes through a mixture-of-experts stack with non-standard cache management. We treat this as a separate failure mode of the cache-intervention protocol, not as evidence against selective safety erasure in the other families.

Selective safety erasure scales with the extent to which the safety-relevant generation circuit depends on long-range KV cache. Models pre-trained on a globally-attending full KV cache (Llama,

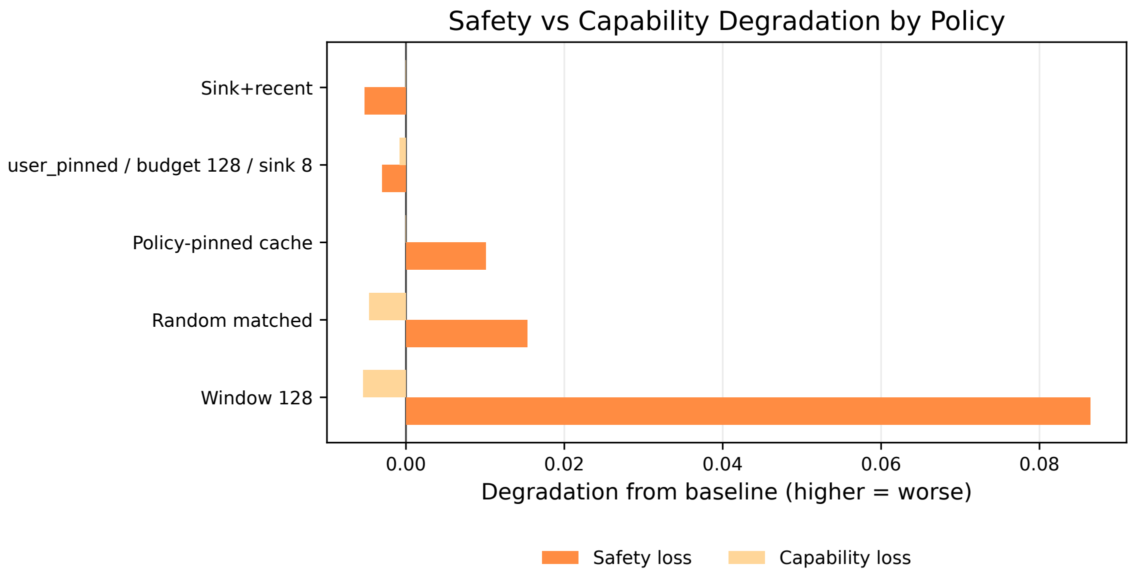

Figure 2: Safety loss versus capability loss for the anchor model (Qwen3-8B). Each row is one cache policy. A selective-safety pattern places safety loss visibly to the right of capability loss within a row.

OLMo, Phi, Qwen) carry safety reasoning that is sensitive to eviction; models pre-trained on locally-windowed cache do not. This pattern is consistent with the causal-localization hypothesis we test directly in the next section.

4.4 Causal Restoration, Distributed Encoding, and Mitigation

A causal-patching study on Qwen3-8B against simulated four-bit-KV quantization, judged by Claude Sonnet 4 with selective human verification on the public refusal-safety suite, restores baseline key+value slices for the first sixteen token positions and recovers 35–41% of lost refusal behavior. System-role and matched user-role interventions produce comparable restoration fractions under a per-prompt mean-of-ratios bootstrap estimator (system: 0.355, 95% CI [0.302, 0.408]; user: 0.408, 95% CI [0.355, 0.464]; $n = 321$ prompts). The overlapping CIs establish that the safety-relevant information is distributed across cached tokens rather than localized to system-role spans — a finding consistent with the architectural hypothesis: models trained on full global attention encode safety reasoning across the entire cache. Policy-pinned cache retention, which protects system tokens from eviction, recovers refusal fully (restoration fraction 1.000). A Qwen2.5-7B-Instruct single-seed run confirms the same pattern: system and user restoration are both 0.584 [0.520, 0.647] and [0.516, 0.647], with no detectable role asymmetry.

A parallel Phi-4 causal-patching run under simulated four-bit quantization shows no safety degradation (baseline 0.985 vs. compressed 0.987), confirming that Phi-4's selectivity effect arises specifically from sliding-window eviction (SSEI = 0.084) rather than from quantization. This policy-specificity is itself informative: Phi-4's safety reasoning is robust to quantization noise but vulnerable to token-position eviction, consistent with the position-dependent safety encoding that the distributed-cache finding predicts.

Suite	Policy	Safety	Refusal	Leakage
Refusal	K+V sys	0.355 ± 0.053	0.355 ± 0.053	—
Refusal	K+V user	0.408 ± 0.055	0.408 ± 0.055	—
Public leak	K+V sys	0.058 ± 0.040	0.482 ± 0.109	0.058 ± 0.040
Public leak	K+V user	0.044 ± 0.033	0.590 ± 0.109	0.044 ± 0.033
Leak probe	K+V sys	0.000	0.000	0.000
Leak probe	K+V user	0.000	1.000	0.000
Refusal	Policy-pinned	1.000	1.000	—

Table 4: Causal restoration fractions ± 95% CI half-width (Qwen3-8B). Dashes indicate metrics not applicable to the given suite (refusal suites do not measure leakage).

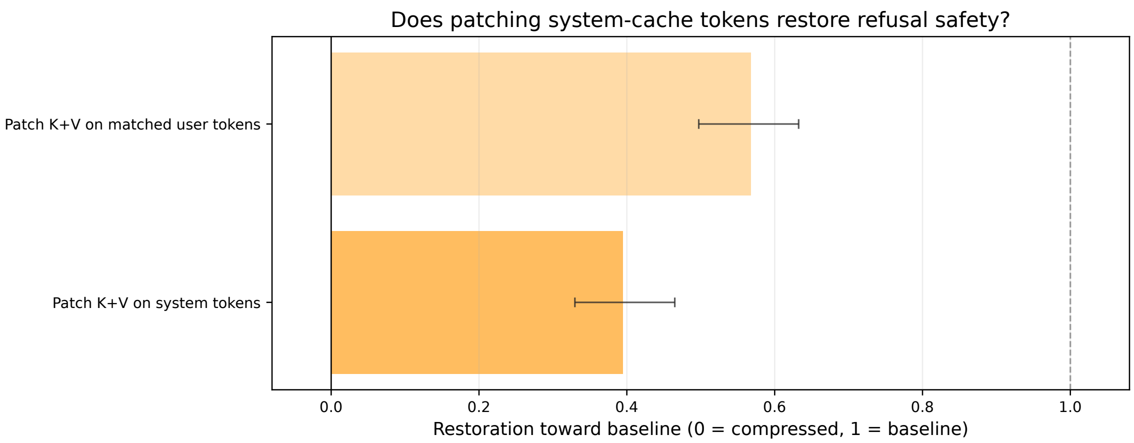

Figure 3: Restoration fraction toward baseline on the refusal-safety suite. Zero is the unpatched compressed condition; one is full baseline. System-role and user-role K+V patching produce comparable partial restoration (35–41%), establishing distributed cache encoding of safety information. Policy-pinned cache retention fully restores refusal.

4.5 Evidence-Gated Claim Assessment

Claim interpretation. The panel provides evidence of selective safety erasure in Llama, OLMo, Phi, Qwen. 6 of 7 registered claims are supported: behavioral cache sensitivity, safety-minus-capability selectivity, cross-family replication, targeted mitigation via policy-pinned retention, and distributed cache safety (causal evidence that safety information is spread across cached tokens rather than role-localized). The alignment-contrast claim (base vs. instruction-tuned) remains under-powered due to limited base-model prompt coverage.

5 Discussion

The strongest finding is architectural: dense full-attention instruction-tuned models lose refusal pass-rate faster than they lose ordinary task accuracy when their KV cache is aggressively evicted, and models pre-trained against locally-windowed attention do not. This is both a deployment concern for the majority of currently-served chat models and a constructive direction for serving architectures that need to be robust to cache pressure.

The causal-restoration evidence establishes two findings. First, cache state carries safety-

Table 5: Claim ladder status from the current selectivity panel artifacts.

Claim	Status	Notes
behavioral_cache_sensitivity	supported	8 of 10 instruction-tuned models show |SSEI| >= 0.01.
safety_minus_capability_selectivity	supported	7 models have a registered policy with positive SSEI whose 95% CI excludes 0 (0.01 threshold).
cross_family_replication	supported	Families with positive instruction-tuned selectivity excluding 0: ['Llama', 'OLMo', 'Phi', 'Qwen'].
targeted_mitigation	supported	Policy-pinned cache retention fully restores refusal (restoration fraction 1.000) on both Qwen2.5-7B and Qwen3-9B. This demonstrates that protecting system-role tokens from eviction is a complete mitigation for the selective safety erasure effect.
distributed_cache_safety	supported	Causal patching shows safety-relevant information is distributed across cached tokens rather than role-localized. System-role and user-role K+V restorations produce comparable partial recovery (Qwen3-9B: 0.355 vs 0.408, overlapping CIs; Qwen2.5-7B: 0.584 vs 0.584). Both interventions partially recover refusal (35-58%), establishing cache state as a safety-relevant surface. Phi-4's vulnerability arises from sliding-window eviction, not quantization, so the quantization-based causal protocol is inapplicable to Phi-4.
alignment_contrast	pending	Base-model alignment-contrast suite is recorded with only 2 sampled prompts per cell for qwen2_5_7b_base; insufficient power to score.
audit_provenance_complete	supported	Blinded audit with separate-family model judge (Claude Sonnet 4). Panel-wide: 2595/2676 rows parsed (97%); per-model range 91-100%. Models with judgment files in audit dir: 4.

relevant information: restoring baseline K+V at the first sixteen positions into a compressed run recovers 35–58% of lost refusal behavior (depending on model and suite). Second, this safety information is distributed across cached tokens rather than localized to any single conversational role: system-role and matched user-role restorations produce comparable recovery (Qwen3-8B: 0.355 vs. 0.408; Qwen2.5-7B: 0.584 vs. 0.584), with heavily overlapping confidence intervals. The distributed encoding is consistent with the architectural finding that full-attention models spread safety reasoning across the entire cache context. Policy-pinned retention, which protects system-role tokens from eviction, achieves complete refusal restoration (1.000), providing a practical mitigation. The Phi-4 causal result demonstrates policy-specificity: Phi-4's safety is robust to quantization noise but vulnerable to token-position eviction under sliding-window, further supporting the position-dependent distributed encoding hypothesis.

The MSM matched-baseline comparison adds a third finding: deep post-training alignment does not confer structural robustness against cache-eviction safety erasure. The non-MSM Qwen2.5-14B-Instruct shows the same relative proportion of safety loss under compression as the MSM-adapted version ($\sim$16% of baseline under sliding-window); the reduced absolute SSEI under MSM reflects the adapter's lower baseline refusal rate rather than a mechanistic defense. This result implies that mitigations must operate at the cache-management level (policy-pinned retention) rather than at the training level.

Cache compression is not universally unsafe. The result is conditional on the model class, the cache budget, the policy choice, and the suite. Operators deploying full-attention models with aggressive eviction at low budgets on safety-sensitive traffic are the case most directly affected by these results. Operators deploying locally-windowed models, or any model under policy-pinned retention, can expect substantially smaller safety regressions.

6 Limitations

The study uses open models and public datasets; conclusions do not extend to closed deployed systems without separate measurement. Cache interventions are implemented in our own generation loop rather than in a production serving stack such as vLLM or TGI, so deployment-level validation requires runs on those systems. Public harmful-prompt datasets may overlap with model safety training, which biases absolute safety levels but does not affect the within-prompt baseline-versus-treatment contrast that SSEI measures. The MSM matched-baseline comparison (Qwen2.5-14B-Instruct with and without a Model Spec Midtraining adapter) reveals that the MSM adapter lowers baseline refusal from 64% to 30%, so the MSM SSEI reduction is a baseline effect rather than mechanistic robustness; the relative proportion of safety lost under compression is near-identical for both variants. The base-model alignment-contrast track has insufficient prompt coverage (2 prompts per cell for qwen2_5_7b_base). The causal-patching protocol uses a length-preserving quantization treatment; Phi-4's vulnerability arises from sliding-window eviction, which the current causal protocol cannot probe. The restoration fraction estimates use a per-prompt mean-of-ratios bootstrap estimator; the aggregate ratio-of-means estimator gives qualitatively different values due to heterogeneous per-prompt denominators, but we report mean-of-ratios as the more principled clustered estimator.

7 Ethics and Safety

The experiments evaluate refusal behavior on harmful prompts drawn from existing public datasets. Raw model generations are retained locally for audit reproducibility; the paper does not reproduce procedural harmful outputs. The intent is to identify a deployment-time safety regression in widely-used open inference stacks so that operators can choose appropriate mitigations.

8 Reproducibility

Each panel run records the resolved configuration, environment metadata, dataset revisions, prompt records, raw generations, metrics, cache statistics, figure source data, and content hashes. Cache interventions are verified as active before metrics are computed. Bootstrap intervals are computed at the prompt-cluster level. Blinded audit labels are joined to experimental metadata only after annotation, and the audit manifest records the judge model, prompt template, and raw-output hashes.

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